

Special Discrete Distributions

ECON 97: Economic Toolkit

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<https://join.iclicker.com/RDNG>

Quiz

Flip a coin 10 times. What is the probability of getting exactly 7 heads?

- A. 0.001
- B. 0.117
- C. 0.234
- D. 0.5

Review: PMF & CDF

- **PMF**: $P(X = x) = f(x)$
 - E.g., $f(3) = P(X = 3)$
 - $\sum_{S_X} f(x) = 1$
 - Support of X (S_X): all possible values of X
- **CDF**: $P(X \leq x) = F(x)$
 - E.g., $F(2) = P(X \leq 2) = f(0) + f(1) + f(2)$
- **Mean**: $\mu = E[X] = \sum_{S_X} xf(x)$
- **Variance**: $\sigma^2 = E[X^2] - E[X]^2$

Special Case: Binomial Distribution

Some experiments have 2 possible outcomes (Pass/Fail, 0/1, Heads/Tails)

- Repeat for many “trials” (n times) without replacement
- Fixed probabilities p (“success”) and $1 - p$ (“failure”)
- $X = \#$ of successes after n trials

$$X \sim \text{bin}(n, p), \quad x = 0, 1, \dots, n$$

$$f(x) = {}_n C_x (p)^x (1 - p)^{(n-x)}$$

$$E[X] = np \quad \text{Var}(X) = np(1 - p)$$

Binomial Distribution - Coin Flip Example

E.g., flip a coin 5 times. What is the probability of exactly 3 heads?

- Heads = success; Tails = failure
- $p = 0.5$ and $1 - p = 0.5$
- $n = 5$

$$P(X = 3) = {}_5C_3(0.5)^3(0.5)^2 = 0.3125$$

$$E[X] = 5 * 0.5 = 2.5$$

$$Var(X) = 5 * 0.5 * 0.5 = 1.25$$

Binomial Distribution - Practice

According to a poll approximately 70% of Americans believe the IRS abuses its power. Let X equal the number of people who believe the IRS abuses its power in a sample of $n = 25$ Americans. Assuming that the poll results are still valid, find the probability that:

- X is at least 13
- X is at most 11
- X is equal to 12
- Give the mean, variance, and standard deviation of X

Binomial Distribution - Computation

Usually compute binomial distribution probabilities using a CDF table:

<i>n</i>	<i>x</i>	<i>p</i>									
		0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
2	0	0.9025	0.8100	0.7225	0.6400	0.5625	0.4900	0.4225	0.3600	0.3025	0.2500
	1	0.9975	0.9900	0.9775	0.9600	0.9375	0.9100	0.8775	0.8400	0.7975	0.7500
	2	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
3	0	0.8574	0.7290	0.6141	0.5120	0.4219	0.3430	0.2746	0.2160	0.1664	0.1250
	1	0.9928	0.9720	0.9392	0.8960	0.8438	0.7840	0.7182	0.6480	0.5748	0.5000
	2	0.9999	0.9990	0.9966	0.9920	0.9844	0.9730	0.9571	0.9360	0.9089	0.8750
	3	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
4	0	0.8145	0.6561	0.5220	0.4096	0.3164	0.2401	0.1785	0.1296	0.0915	0.0625
	1	0.9860	0.9477	0.8905	0.8192	0.7383	0.6517	0.5630	0.4752	0.3910	0.3125
	2	0.9995	0.9963	0.9880	0.9728	0.9492	0.9163	0.8735	0.8208	0.7585	0.6875
	3	1.0000	0.9999	0.9995	0.9984	0.9961	0.9919	0.9850	0.9744	0.9590	0.9375
	4	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

E.g., for $n = 3$ with probability of success $p = 0.1$:

$$F(2) = 0.999 = P(X \leq 2)$$

Binomial Distribution - Practice

It is claimed that 15% of ducks in a particular region have patent schistosome infection. Suppose that seven ducks are selected at random. Let X equal the number of ducks that are infected.

- Assuming independence, how is X distributed?
- Find $P(X = 1)$, $P(X \geq 2)$, and $P(X \leq 3)$

Binomial Distribution - Practice

A national study showed that approximately 45% of college students binge drink. Let X be the number of students in a random sample of size $n = 12$ who binge drink. Find the probability that

- X is at most 5
- X is at least 6
- X is equal to 7
- Give the mean, variance, and standard deviation of X

Binomial Distribution - Practice

It is believed that 20% of Americans do not have any health insurance. Suppose this is true and let X equal the number with no health insurance in a sample of $n = 15$ Americans.

- How is X distributed?
- Give the mean, variance, and standard deviation of X
- Find $P(X \geq 5)$

Special Case: Poisson Distribution

If we count the number of events over a span of time/space:

- Average rate of occurrence λ
- $X = \#$ of occurrences in a given time/area

$$X \sim \text{Pois}(\lambda), \quad x = 0, 1, 2, \dots$$

$$f(x) = P(X = x) = (e^{-\lambda}) \frac{\lambda^x}{x!}$$

$$E[X] = \text{Var}(X) = \lambda$$

Binomial: $\#$ of successes in n trials

Poisson: $\#$ of successes in a period of time

Poisson Distribution - Example

A store receives 30 customers per day on average. What is the probability that it receives exactly 25 customers tomorrow?

- $\lambda = 30$
- $x = 25$

$$f(25) = (e^{-30}) \frac{30^{25}}{25!} \approx 0.051$$

$$E[X] = 30 = Var(X)$$

Poisson Distribution - Computation

Can also use a CDF table for Poisson probabilities:

	$\lambda = E(X)$									
x	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
0	0.905	0.819	0.741	0.670	0.607	0.549	0.497	0.449	0.407	0.368
1	0.995	0.982	0.963	0.938	0.910	0.878	0.844	0.809	0.772	0.736
2	1.000	0.999	0.996	0.992	0.986	0.977	0.966	0.953	0.937	0.920
3	1.000	1.000	1.000	0.999	0.998	0.997	0.994	0.991	0.987	0.981
4	1.000	1.000	1.000	1.000	1.000	1.000	0.999	0.999	0.998	0.996
5	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.999
6	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

E.g., for $\lambda = 0.7$:

$$F(2) = 0.966 = P(X \leq 2)$$

Poisson Distribution - Practice

A website receives customer support chats at an average rate of 3 chats per hour. Assume the number of chats received follows a Poisson distribution. Let X be the number of chats received in one hour.

- What is the probability that exactly 2 chats are received in the next hour?
- What is the probability that no chats are received in the next hour?
- What is the probability that at least 1 chat is received in the next hour?